

ERRATA

The errors of this project, collected: what was wrong, how it was found, what it cost, and what it became.

Every measuring instrument in this repository exists because a specific mistake demanded it. The mistakes are scattered across seven papers as dated corrections; this paper gathers them in one place, because a correction that has to be hunted for protects nobody. The rule for inclusion: an error is listed if it changed a published number, a published claim, or the instrument. Typos and workflow accidents are not science and are not here.

0x44 Zero Systems · Author: 0x44 DANTE · As of: 1 September 2026

Every number here was re-counted when this paper was written, from the file it came from, and every one of those files is published: the repository holds them, except the 92,091 campaign logs of section 4, which are attached to release v1.0 as an archive. One entry rests on evidence that is not ours and says so where it stands: section 6 counts Leitner-Ankerl's tables, which are his.

1 THE JUDGE THAT EXITED SILENTLY

What was wrong. PractRand 0.94, built with modern GCC at -O3, has a function missing its return statement. Undefined behaviour: the process ends silently after the first checkpoint, and every mixer — including deliberately broken ones — reads “clean to 2^{10} ”.

Whose finding this is: not ours. The missing return was reported in 2019 with a patch (SourceForge bug #12), the author confirmed the cause in his own forum, and upstream removed it in pre0.95 by declaring the function `void`. Release 0.96 of December 2025 has no such bug — downloaded and read on 1 September 2026. Ours is only the shape it takes under TDM-GCC 10.3 at -O3, which none of those sources describes: not a segfault but a silent exit with code 0, leaving a log that reads like a clean pass.

How it surfaced here. Through a contradiction with the literature: mix13 appeared to pass, and Evensen publishes it failing between 2^{16} and 2^{22} . An instrument agreeing with itself while disagreeing with the world is how every error in this paper surfaced, and this was the first.

What it became. The documented build patch, and the rule the whole rig now runs on: nothing is measured until the rig has convicted a known-guilty mixer in the same run. Full analysis with upstream history: `rrc/PRACTRAND_FINDING.md`.

2 THE ANCHOR THAT WAS A MIRROR

What was wrong. The rig's upper calibration weight was mfx9 — our own mixer. It behaved impeccably in the slot, and that was exactly the problem: “mfx9 holds” only ever said the rig agreed with yesterday's rig. A weight is only a weight if somebody else weighed it.

What it became. NASAM took the slot on 29 August 2026: Evensen publishes it passing RRC-64-42, so agreement with it means something outside this rig. The same fallacy was later found once more, in the most expensive place: the meer10 diploma's upper anchor was meer10 itself holding its own gauntlets, and the README now says so beside the headline rather than three pages down.

3 THE ANCHOR THAT ERASED ITS OWN EVIDENCE

What was wrong. The diploma's anchor re-measured the same four gauntlet rotations into the same filenames on every restart, and overwrote four finished 1 TB diploma stream logs. The rig built to protect evidence destroyed it.

What it cost, and what it became. Four streams re-measured from scratch on 28 August (all four returned clean to 2^{40} , identical to their checkpoints). Anchor runs carry their own filename prefix since. Full account: section 7 of *The Calibration Weights*.

4 THE WATCHDOG BUILT FOR THE WRONG SHAPE

What was wrong. The stall detector asked whether a stream had produced a *first* checkpoint. The stall it was named after freezes at about 8 KB — four checkpoints in. For its entire existence the detector never fired on the failure it existed to catch, and its only anchor was the one that proves it spares the healthy.

What it cost. Counted afterwards on the measuring machine, over the 92,091 stream logs of the finished cost-5 campaign: 1,337 stall aborts and 72 timeouts, each of those streams paying two full watchdog timeouts to produce nothing before the fix. Until 1 September this was the one measurement behind this paper a reader could not recount, on the grounds that 92,091 files were too large to publish. That was a packaging problem stated as a size problem: the files do not belong in a git tree, because every clone would pay for them, but they pack to 10.7 MB. They are attached to release v1.0 as `k5_campaign.tar.gz`, and the count above is now yours to check.

What it became. A detector that tracks output growth, a floor measured down from 20 s to 8 s, and the third anchor pair: a deliberately frozen feeder **MUST** be cut (THE SLEEPER) and a healthy stream **MUST** survive (THE PATIENT). A check that has only ever been seen to pass has not been seen to work.

5 CLAIMS WITHOUT FILES

The worst class of error in this repository, because each one looked exactly like a result. Four documents stated measurements that no file anywhere recorded:

Claim	Reality	Resolution
meer10 passed the 2 ³⁴ depth stage, 64/64	no run existed; the candidate list of the 19 August depth run holds 23 bswap chains and nothing else	run on 28 Aug: 64/64 clean, results/depth_meer10_*
lu9 passed the same stage, 64/64	no file in any version of the repository or the private rig	run on 30 Aug: 64/64 clean, results/depth_lu9_*
six smoke figures (mfx9, meer10, mfa9 hold 32/32; mulfold2 falls 11/32, mf2b 8/32, moremur 16/32)	quoted in the papers with no published stream logs	re-measured 30 Aug: every figure reproduced exactly, results/smoke_evidence/
raw logs withheld because large runs come to “hundreds of megabytes”	the whole 256 TB diploma comes to 1.1 MB of logs	256 diploma logs published; 3,870 raw logs, 7.1 MB, in results/

The pattern behind all four: a cascade table describes what is supposed to happen, and after enough re-readings it reads as what did happen. Every claim turned out to be true when measured. That is luck, not vindication: a true claim about an unperformed measurement is still a claim, and the runs were performed because of what it would mean if one had come back false.

6 NUMBERS THAT MEANT SOMETHING ELSE

Published	It actually was	Where corrected
the diploma took 39.5 hours	the last of at least three segments. The earliest surviving log opens on 22 Aug with 20 streams already booked, so the campaign began before that and ended on the 28th; no total was kept. The cause lived in the code: a per-process clock paired with a whole-campaign stream count	in every text. The tool now writes hours_this_segment; the published result files predate the rename and keep their own field names
the entry test runs “all disguises”	8 of 64 rotations. In the depth data, 23 stream deaths land below the entry test's own volume — every one on a rotation it never runs	README and FAMILY state the sampling and the measured gap
Leitner-Ankerl saw two AES mixers die at 2 ¹³ –2 ²⁰	his aes2 dies at 2 ¹³ –2 ¹⁶ ; his aes3 says “only tested up to -tmax 20” and does not fail there in 124 of 128 cells. A test ceiling was cited as a death	COSTLIMIT, with the source counted line by line
mixers were compared across cascade stages	the score is a minimum over streams and falls as more are taken: 2.5 doublings on average between stages, 13 in the worst case, measured on the 23-chain family	README method section; comparisons are per-stage now

Published	It actually was	Where corrected
27 of 30 null-run streams were fully clean	28; miscounted once, caught by a reader re-counting	results/README

7 THE METRIC THAT HAD NEVER MET A CLOCK

What was wrong. Every cost figure in this repository comes from a weight table — and until 31 August 2026 nothing had been measured against it. The quality axis carried 256 terabytes; the cost axis carried a convention. Worse, the table prices two structurally identical operations differently (double shift-xor at 2, double rotate-xor at 3), and the headline sentence “entry and diploma are one cost point apart” hung on exactly that unpublished entry.

What the measurement said. A dependent-chain benchmark (`rrc/bench_cost.cpp`, 100 million iterations, three runs) disagreed with the table twice: the classical shift-multiply mixers are overpriced, and `meer10` at cost 10 runs slower than `mix13`, `fmix64` and `moremur` at cost 12 — 3.28 ns against 3.01–3.07 on the measuring machine. The repository's front page now carries the number that goes against it. What survives is `meer10` against NASAM, 3.28 against 3.84, and the headline sentence was rewritten as `structure` (same skeleton, different middle member) so it no longer depends on any weighting.

8 DISCOVERIES THAT WERE NOT

Three findings were announced inside working sessions and killed by their own control measurements before publication. They are listed because the controls are the method:

The test-distribution structure. K5 chains seemed to die to characteristic tests per depth — until the same counting was run on the K9 population at equal depth: the pattern is `PractRand`'s own test schedule (6 tests at 2^{10} , 24 at 2^{12} , 47 at 2^{13}), not a property of the class.

The folding share. An apparent relationship between mixer quality and the share of folded-stream detections dissolved under the same control.

Two readings of the K5 map. “Every deep skeleton carries a fold” and “the bottom block ends in permutations” — both read off the picture, both wrong when counted (six of nine deep skeletons carry a fold; the bottom block is about missing back-transport, a sharper finding than either guess). The map stayed; the sentences were replaced by counts.

9 THE BASELINE NOBODY HAD MEASURED

What was wrong. “At terabyte volumes a run with no suspicious at all would itself be suspect” — a published sentence about a null distribution nobody had measured.

What the measurement said. Thirty 2 GB streams from three sources whose randomness is not in dispute (`chacha8`, `sfc64`, `xsm64`), same judge, same flags: 3 suspicious lines, 0 FAIL, 28 of 30 fully clean. The sentence was wrong at that depth and was replaced by the numbers (`results/null_run_result.json`). At diploma depth the baseline remains unmeasured, and the 34 suspicious events across 256 TB stand without one — written down as such.

10 WHAT THE LIST BECAME

Each class of error above hardened into a mechanism, and the mechanisms are the actual product of this project:

Error class	Mechanism it became
a judge that exits silently (1)	the build patch, plus THE GUILTY: a known-bad mixer must fall in the same run
self-calibration (2)	THE INNOCENT must be externally published; anchors are weights only if somebody else weighed them
evidence destroyed by the rig (3)	anchor runs write under their own prefix; one results folder per campaign
silent verdicts (4, 5)	THE MISTRIAL and THE ADJOURNED: unmeasured and aborted streams must say so; absence of a FAIL is never a pass
a watchdog never seen biting (4)	THE SLEEPER and THE PATIENT: both directions, or neither
claims without files (5)	provenance blocks in the result files a measuring command writes, from 30 August onwards — eight of them carry one, and the two that should and do not are named in results/README.md; every published number re-counted against its file before release
unmeasured conventions (7, 9)	the clock benchmark and the null run; a convention that survives its own measurement is worth more afterwards

Two things this paper is not. It is not a confession — every entry earned its place by changing a number, a claim or the instrument, and entries that changed nothing were left out. And it is not complete by construction: it lists the errors that were caught, and the ones that were not are, by definition, still in the other papers. The instrument that found most of these was an adversarial re-reading of our own files; the same reading applied again will find more.

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As of: 31 August 2026. Rig and method: see *The Ladder*. The anchor incidents in full: see *The Calibration Weights*. The evidence files named here live under results/, except two that live under rrc/ (the PractRand finding and the cost benchmark) and the two unpublished sources named on the front page.